

PROJECT REPORT

INFO STEALER

DETECTION & RESPONSE

Integrated SIEM & SOAR Project

Information Security Analyst — Project Report

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1. Introduction

1.1 Idea

The core concept is to emulate an Information Stealer malware sample in a controlled environment, capture its behavioral logs, and build a semi-automated Security Operations Center (SOC) pipeline. This involves detecting the threat through a SIEM solution, forwarding the resulting alerts to a SOAR platform, and following the Incident Response (IR) steps for containment and eradication.

1.2 How It Is Applied

- Executing the Info Stealer safely to generate telemetry.
- Ingesting logs into Elastic Security (SIEM).
- Creating custom detection rules to detect the Info Stealer.
- Utilizing a custom Python service to pull alerts and push them to a Shuffle SOAR webhook.
- Automating threat-intelligence gathering and alerting through SOAR playbooks.
- Sending an automated email alert to the analyst whenever a new detection is triggered.

1.3 How It Helps in Cybersecurity Defense

It drastically reduces the Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR), both of which directly affect the SLA. By automating Tier 1 SOC analyst tasks — such as IP reputation checking and alerting — the defense team can focus on complex threat hunting rather than manual triage. Note that this automation covers detection, enrichment, and alerting only; the actual response and containment decisions are still made and carried out by the analyst team.

1.4 Add-Ons

Threat-intelligence enrichment using the VirusTotal API v3, integrated directly into the automated alerting playbook.

1.5 Future Additions

Integrating active containment mechanisms — such as pushing API commands to edge firewalls or EDR platforms (e.g., Windows Defender, CrowdStrike) — to automatically isolate an infected host from the network without human intervention. This is a planned enhancement only: containment today is performed manually by the response team, not by the pipeline itself.

2. Methodology

2.1 Technology Stack

TOOL	ROLE	FUNCTION
Elasticsearch	SIEM — Data Store	Indexes and searches all ingested log events.

TOOL	ROLE	FUNCTION
Kibana	SIEM — UI & Rules	Detection rule creation, alert management, and dashboards.
Filebeat	Log Shipper	Collects endpoint logs and forwards them to Elasticsearch.
Shuffle	SOAR Engine	Executes automated alerting and enrichment playbooks triggered by Kibana alerts.
Python 3	Scripting Engine	Bridges the SIEM and SOAR platforms using the requests and datetime libraries for API integration.

2.2 Timeline

PHASE	ACTIVITY	PERIOD
Setup	Environment design & tool installation	February
Ingestion	Log pipeline configuration & data validation	March
Detection	Kibana rule development & dashboard creation	March
Responding	Shuffle integration & playbook development	April
Testing	End-to-end simulation with Atomic Red Team	April – May
Docs	Report writing & final review	May

2.3 Implementation

The idea was implemented by configuring a continuous background service (`elastic_to_shuffle.py`) that queries the `internal.alerts-security.alerts` index in Elastic every 60 seconds. Upon detecting a signal event, the script extracts the raw JSON payload and posts it to a Shuffle webhook. Shuffle then extracts the malicious IP, queries VirusTotal, and uses conditional logic (`malicious > 0`) to trigger an alert containing the extracted IOCs — which the analyst then reviews and acts on manually.

3. Proposed Impact

3.1 Objectives

- Successfully simulate and detect advanced malware behavior.
- Establish a zero-touch alert-forwarding mechanism from SIEM to SOAR.
- Build a functional OSINT automation module inside the SOAR platform.

3.2 Percentage of Achievement

Approximately 85–90%. The detection, ingestion, and enrichment pipelines are fully operational. The remaining percentage accounts for automated eradication and isolation scripts, which are planned but not yet implemented — containment today is still carried out manually by the response team.

3.3 Difficulties Faced

- Handling API pagination and authentication headers.
- Formatting complex JSON payloads to be accepted by webhooks.
- Resolving dynamic data-mapping errors inside the SOAR platform (e.g., translating raw variables such as \$exec.source.ip into actual IP addresses for third-party APIs).
- Configuring cloud vs. local API keys for SMTP/email routing.

3.4 The Future Adds

Developing automated eradication playbooks that would delete malicious payload hashes from the endpoint via EDR integration — a planned future capability, not part of the current implementation.

4. Results

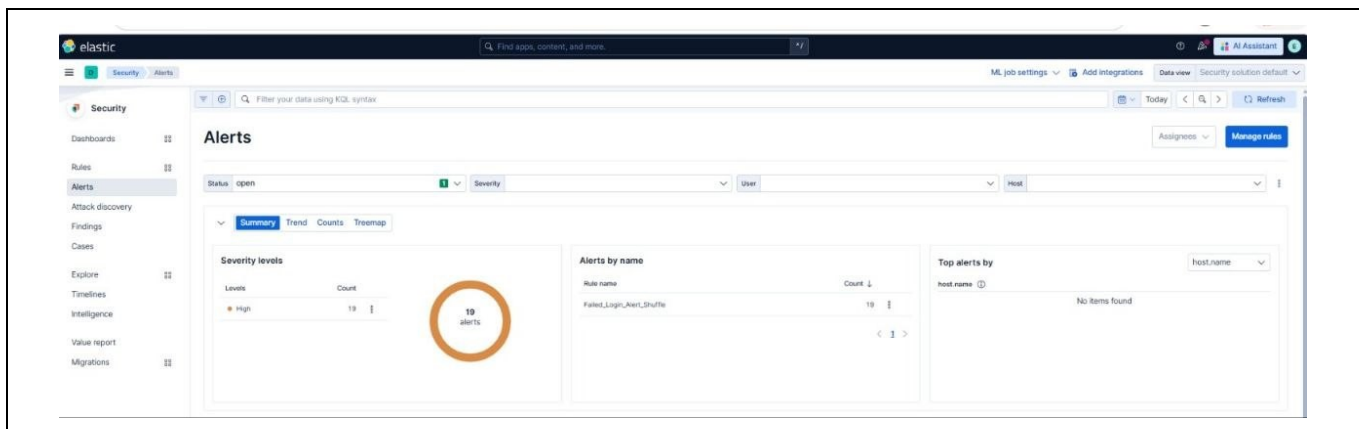


Figure 1 — Elastic SIEM showing the triggered alert.

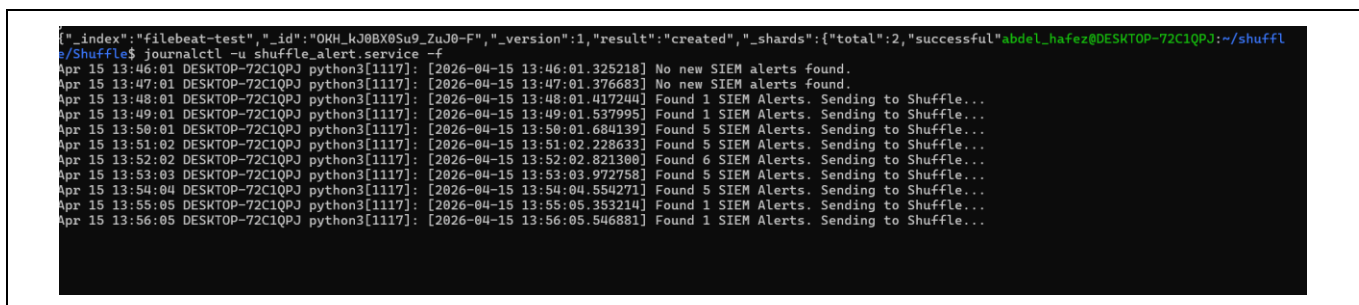


Figure 2 — Terminal output showing the Python script locating and forwarding SIEM alerts.

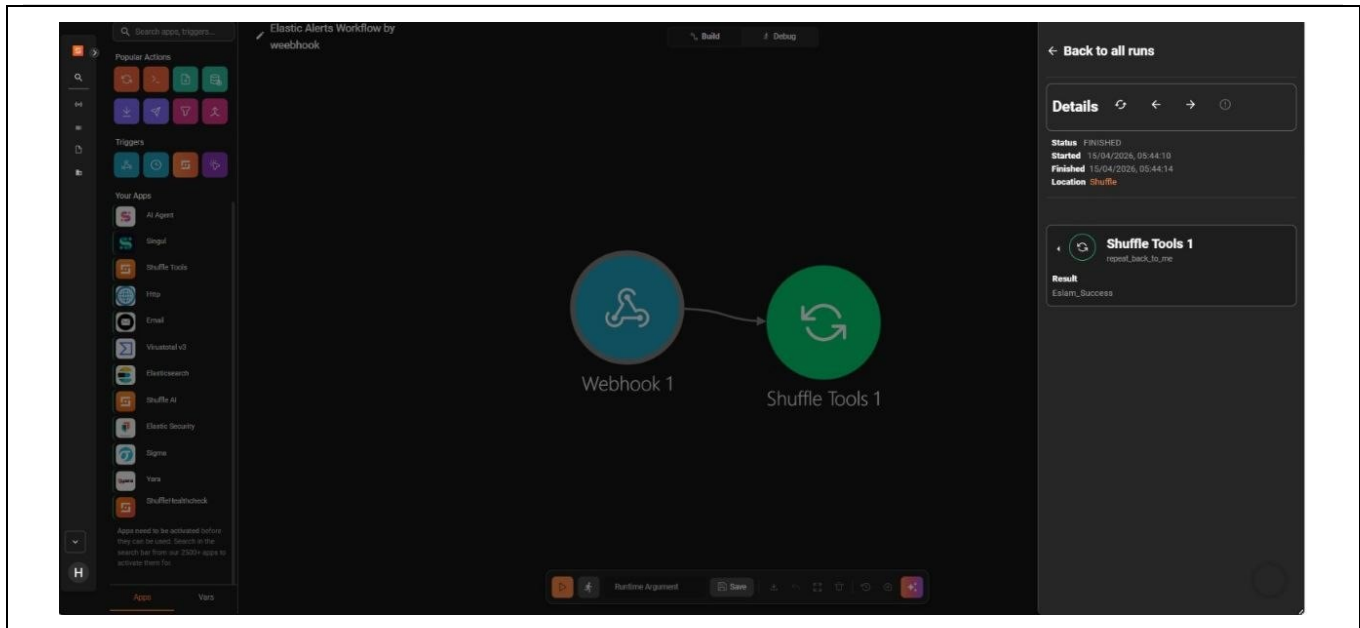


Figure 3 — The Shuffle workflow illustrating the error-handling path prior to resolution.

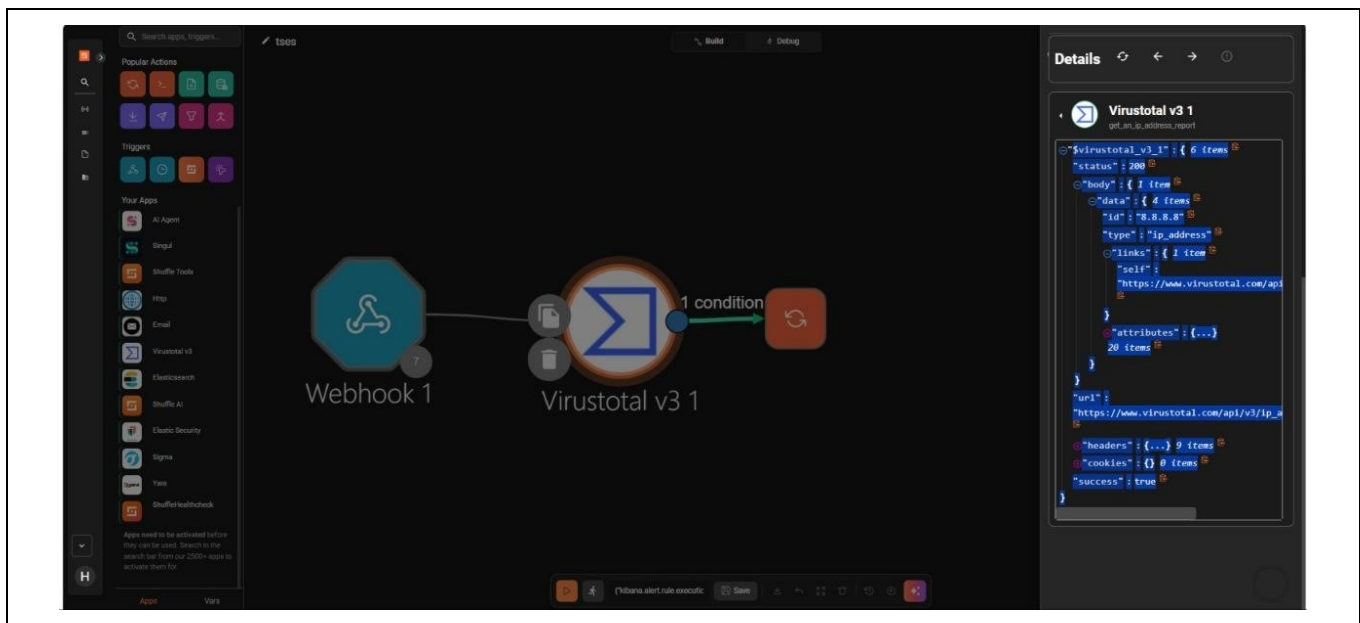


Figure 4 — The Shuffle workflow showing the successful execution path through the VirusTotal condition.

```

WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.0.7:5000
Press CTRL+C to quit
127.0.0.1 - - [21/Apr/2026 00:28:51] "GET / HTTP/1.1" 404 -
127.0.0.1 - - [21/Apr/2026 00:28:51] "GET / HTTP/1.1" 404 -
127.0.0.1 - - [21/Apr/2026 00:29:05] "GET / HTTP/1.1" 404 -
-----
[PredatorSec Active Response]
Time: 2026-04-21 00:30:52
Action: BLOCK IP
Target IP: 194.26.135.119
Attack Type: PredatorSec_SOAR_Python_Test
Status: Rule added to Local Firewall successfully.
127.0.0.1 - - [21/Apr/2026 00:30:52] "POST /api/firewall/block HTTP/1.1" 200 -
-----
[PredatorSec Active Response]
Time: 2026-04-21 00:33:57
Action: BLOCK IP
Target IP:
Attack Type:
Status: Rule added to Local Firewall successfully.
127.0.0.1 - - [21/Apr/2026 00:33:57] "POST /api/firewall/block HTTP/1.1" 200 -

```

Figure 5 — Consecutive automated executions blocking the target IP.

4.1 Live Product

A fully functional, integrated SIEM-to-SOAR pipeline capable of executing automated detection, enrichment, and email-alerting playbooks within seconds of a SIEM trigger. Incident response and containment actions are reviewed and carried out manually by the SOC team; they are not automated by the pipeline.

4.2 Evidence of Achievement

Successful HTTP 200 OK responses from VirusTotal when parsing dynamic IPs, and time-stamped logs proving the automated flow of data from detection through to email notification for the analyst.

4.3 Cyber Kill Chain Mapping

- Delivery / Installation: Info Stealer execution.
- Command & Control (C2) / Actions on Objectives: malware attempting to exfiltrate data, caught by the SIEM detection rules.

4.4 MITRE ATT&CK

- TA0001 — Initial Access: monitored via rules detecting suspicious process executions or unauthorized access.
- TA0010 — Exfiltration: monitored via rules detecting anomalous outbound network traffic to known-bad IPs.

5. Executive Summary

This project successfully demonstrated a proactive, semi-automated defense workflow against Information Stealer malware. By integrating Elastic SIEM with Shuffle SOAR, the detection of Initial Access and Exfiltration behavior was transformed from a manual monitoring task into an automated detection-and-alerting pipeline — while the incident response and containment steps themselves remained manual, carried out by the analyst team once notified. The system efficiently ingested logs, triggered alerts, and used a custom Python bridge to initiate each SOAR playbook. Despite technical challenges involving dynamic JSON data mapping, webhook configuration, and API authentication, the pipeline successfully enriched threat data using VirusTotal and generated automated email notifications to

support the team's manual response decisions. The results demonstrate the value of integrating SOAR into modern SOC environments, paving the way for faster detection and more informed, analyst-driven incident response.